

Multiple Choice

1. **Answer:** C

Options: A) Deficiency of leptin caused by a mutation in the leptin gene; B) Lack of leptin release from the adipose tissue; C) High concentrations of leptin without reduction in energy intake; D) Low concentrations of leptin with increased energy intake

Note: Leptin resistance occurs when the body produces high levels of leptin, a hormone that regulates energy balance, but fails to respond effectively to its signals, resulting in continued energy intake instead of a decrease.

2. **Answer:** D

Options: A) Suppresses milk production; B) Alters the composition of breast milk; C) Increases the risk of early cessation of breastfeeding; D) all of the options given are correct

Note: Smoking can affect breastfeeding by reducing milk production, altering the composition of breast milk, and increasing the risk of health issues for both the mother and the infant, making all the given options correct.

3. **Answer:** C

Options: A) BMI; B) DXA; C) Four-compartment model; D) MRI

Note: The four-compartment model is the most accurate method for estimating changes in body fat during weight loss because it separates body mass into fat mass, lean mass, bone mass, and total body water, providing a comprehensive analysis of body composition changes.

4. **Answer:** D

Options: A) fasting blood glucose $\geq$ 140 mg/dl; B) random glucose $>$ 160 mg/dl; C) 2 hour post prandial glucose $\geq$ 126 mg/dl; D) fasting blood glucose $\geq$ 126 mg/dl

Note: A fasting blood glucose level of 126 mg/dl or higher is established as the diagnostic threshold for diabetes by health organizations, indicating impaired glucose metabolism.

5. **Answer:** A

Options: A) Folate; B) Riboflavin; C) Thiamin; D) Vitamin C

Note: Folate is the correct answer because it plays a crucial role in one-carbon metabolism by facilitating the transfer of one-carbon units in various biochemical processes, including DNA synthesis and amino acid metabolism.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Acetaldehyde, a byproduct of alcohol metabolism, is classified as a probable human carcinogen because it can cause DNA damage and promote tumor formation, thereby contributing to cancer development.

7. **Answer:** True

Options: A) True; B) False

Note: Transaminases, which are enzymes that catalyze the transfer of amino groups between amino acids and alpha-keto acids, require pyridoxal phosphate, the active form of vitamin B₆, as a coenzyme to function effectively in these transamination reactions.

8. **Answer:** True

Options: A) True; B) False

Note: Deuterium, a stable isotope of hydrogen, is used in health sciences to measure total body water because it can be easily tracked in biological systems without causing harmful effects.

9. **Answer:** True

Options: A) True; B) False

Note: Food availability and consumption can be measured at five stages within the food supply chain-production, processing, distribution, retail, and consumption-allowing for a comprehensive understanding of how food moves from farms to consumers.

10. **Answer:** True

Options: A) True; B) False

Note: Eating only locally grown food in central Africa can lead to iodine deficiency because the region often lacks iodine-rich foods, such as seafood and iodized salt, which are essential for maintaining adequate iodine levels in the diet.

Long Form Response

11. **Answer:** Alcohol consumption is linked to an increased risk of developing esophageal cancer, primarily due to its effects on the body. When alcohol is consumed, it is metabolized into acetaldehyde, a substance that can damage the DNA in cells lining the esophagus. This damage may lead to mutations that contribute to cancer development. Additionally, alcohol can irritate the esophageal lining, making it more susceptible to other carcinogenic substances, especially when combined with tobacco use. Therefore, reducing alcohol intake can be an important step in lowering the risk of esophageal cancer.

Note: correct because it identifies the key mechanisms by which alcohol contributes to esophageal cancer risk, including the formation of DNA-damaging acetaldehyde and the irritation of the esophageal lining, particularly in the context of concurrent tobacco use.

12. **Answer:** Vitamin B₆ plays a crucial role in the transamination process, which is the transfer of amino groups from one amino acid to another. This process is essential for amino acid metabolism, allowing the body to create non-essential amino acids and convert them into different forms as needed. Vitamin B₆ serves as a coenzyme in this reaction, meaning it helps enzymes carry out their functions effectively. This is significant for overall health because it supports protein synthesis, neurotransmitter production, and the metabolism of carbohydrates and fats. Adequate levels of Vitamin B₆ can help maintain a healthy immune system, improve mood, and support brain function.

Note: correct because it accurately describes Vitamin B₆'s essential role as a coenzyme in the transamination process, which is crucial for amino acid metabolism, protein synthesis, and overall health.